Cross-Event Optical Relation Distillation for SAR Landslide Mapping

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***Abstract*—Optical imagery maps landslides most accurately, but clouds often leave synthetic aperture radar (SAR) as the only rapid observation after a disaster. We use optical data only during training to improve a SAR landslide mapper that receives SAR backscatter and terrain at inference. Landslide inventories are collected event by event, and a SAR network trained on them learns to recognize the training events rather than landslides. Conventional distillation from an optical teacher does not resolve this: the teacher reproduces the training labels, and most optical content is not observable in SAR. We find that students improve when they do not copy the teacher's pairwise region relations. We therefore distill only the distribution of each region's similarities to regions of other disaster events, measured by a closed-form one-dimensional Wasserstein distance, and give the student decoder one prototype bank shared by all events and decoder scales. On the official split of a 23-event Sentinel-1/2 benchmark, the student reaches 56.45% intersection over union (IoU) and 77.59% average precision (AP) with SAR and terrain input only, compared with 52.20% and 72.91% for the same U-Net trained without optical data.**

***Index Terms*—Knowledge distillation, landslide mapping, privileged information, synthetic aperture radar (SAR), Sentinel-2.**

I. Introduction

SAR supports landslide mapping when clouds or darkness prevent optical acquisitions during emergency response [1]–[3]. Convolutional networks map landslides from SAR backscatter [4], [5], yet a recent multi-sensor benchmark shows that optical features remain clearly more accurate than SAR features, and that SAR models degrade most on events not represented in training [6].

Training archives frequently contain co-registered optical imagery even when an emergency map must rely on SAR. Learning with privileged information and knowledge distillation (KD) exploit such training-only inputs [7], [8]. Remote-sensing KD transfers predictions, features, or attention between modalities [9]–[12], and relational KD transfers similarities between samples [13]–[15].

Two properties of landslide data limit these approaches. First, inventories are grouped by triggering event; a SAR U-Net reaches 90.9% IoU on validation patches from its training events but 52.2% on test events. Second, a multimodal teacher fits the training labels almost exactly, so its outputs carry little information beyond the labels, while its features encode optical evidence that SAR cannot reproduce. We observe that students which reproduce the teacher's pairwise region relations more closely generalize worse to new events.

We therefore transfer relational structure rather than content, and only its distribution across events. Our contributions are: 1) a cross-event relation distillation that matches, for each region, the distribution of its similarities to regions of other events with a closed-form Wasserstein distance and no pair correspondence; 2) a prototype bank shared by all events and by three decoder scales, which gives this relational structure a deployable form for landslides of very different sizes; 3) an analysis on a 23-event benchmark showing why output, feature, and pairwise relation transfer underperform for event-grouped landslide data.

II. Method

A. Problem Formulation

Let X_s denote the student input: pre- and post-event Sentinel-1 VV/VH backscatter, elevation, and slope (six channels). The teacher additionally receives pre- and post-event Sentinel-2 bands X_o (twelve channels). Both networks follow the U-Net of [6], [16]. The teacher is trained with the segmentation loss and then frozen; the student is deployed on X_s only. Fig. 1 summarizes the method.

B. Scale- and Event-Shared Prototypes

After decoder stage $l \in \{1,2,3\}$ (16, 32, and 64 pixels), the student feature $f_l \in \mathbb{R}^{C_l \times H_l \times W_l}$ is compared with one bank of K learnable prototypes $\{P_k\}$ shared by all events and stages:

$$a_{lk}(p) = \mathrm{softmax}_k\Big(\cos\big(W_q^l f_l(p), P_k\big)/\tau\Big), \tag{1}$$

$$f_l'(p) = f_l(p) + W_o^l \sum_k a_{lk}(p)\, W_v^l P_k, \tag{2}$$

where W_q^l, W_v^l, and W_o^l are stage-specific 1×1 projections, $\tau = 0.1$, and $K = 16$. W_o^l is zero-initialized, so training starts from the plain U-Net. The prototypes are the only parameters that every pixel of every event reads at every scale, so a concept learned for one landslide size is available to the others; they remain in the deployed network.

C. Cross-Event Wasserstein Relation Distillation

The relation loss acts at the semantic stage $l = 2$ (32×32). Teacher and refined student features there are average-pooled over valid pixels into 4×4 region tokens per patch and normalized to unit length, giving t_i (teacher) and s_i (student, from f_2'). Finer stages are not constrained: at 64×64 a region covers about 160 m and its features mainly reflect speckle and texture, which differ between acquisitions. For region i from event e_i, let $\mathcal{N}_i = \{j : e_j \neq e_i\}$ be the regions of other events in the mini-batch. The teacher and student similarity sets are

$$\mathcal{T}_i = \{t_i^\top t_j\}_{j\in\mathcal{N}_i}, \qquad \mathcal{S}_i = \{s_i^\top s_j\}_{j\in\mathcal{N}_i}, \tag{3}$$

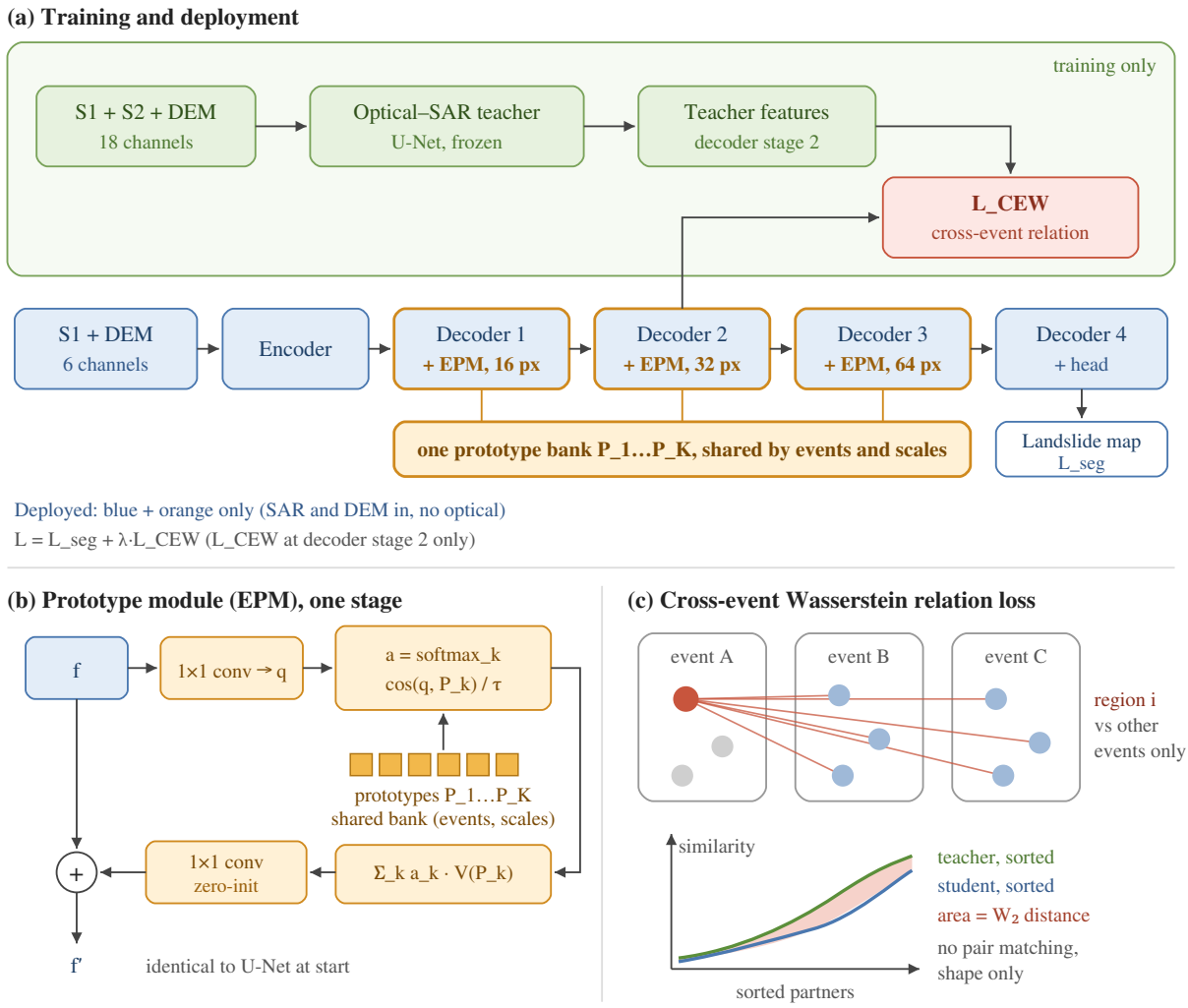


Fig. 1. (a) Training and deployment. The optical–SAR teacher (green) is used only during training. (b) Event-shared prototype module. (c) Cross-event Wasserstein relation loss: each region is compared only with regions of other events; teacher and student similarities are sorted separately and compared as distributions.

each shifted by its minimum over cross-event pairs. Sorting each set separately gives $\tilde{\mathscr{T}}_i$ and $\tilde{\mathscr{S}}_i$, and the squared one-dimensional Wasserstein distance between the two empirical distributions is

$$L_{\mathrm{CEW}} = \frac{1}{N}\sum_i \frac{1}{|\mathscr{N}_i|}\sum_{r=1}^{|\mathscr{N}_i|}\left(\tilde{\mathscr{T}}_i[r] - \tilde{\mathscr{S}}_i[r]\right)^2. \tag{4}$$

Sorting removes pair correspondence: the student decides which regions of other events are close or distant, while the teacher prescribes how the cross-event similarities are distributed. Same-event pairs never enter the loss, so acquisition conditions shared within an event cannot satisfy it.

D. Learning Objective

The student minimizes

$$L = L_{\mathrm{BCE}} + L_{\mathrm{Dice}} + \lambda L_{\mathrm{CEW}}, \tag{5}$$

with binary cross-entropy and foreground soft Dice on valid pixels and $\lambda = 30$, for which the initial gradient of λL_{CEW} is about 20 times that of the segmentation loss. A weight that equalizes the two gradients at initialization lowers IoU to 53.09% in the single-stage configuration.

III. Experiments

A. Dataset and Settings

We use the Sentinel-1/2 landslide reference data of [6] with its official split: 355 training and 88 validation patches from 23 events, and 120 test patches of 128×128 pixels at 10 m. The test set contains 44 patches from four independent events (Ind.) and 76 patches from four dependent events whose training patches are separated by a spatial block split (Dep.). Inputs follow the one-pre/one-post setting of [6]: γ^0 VV/VH in decibels, Copernicus DEM elevation, and slope; the teacher adds Sentinel-2 B2, B3, B4, B8, B11, and B12.

All networks use the U-Net of [6], AdamW with a learning rate of 10^{-4} and weight decay of 0.01, batches of 16, random flips and rotations, and 200 epochs (constant rate for 100 epochs, then cosine decay). Validation IoU every 20 epochs selects the checkpoint; each model is tested once after selection, with threshold 0.5. All students share one initialization

TABLE I
TEST RESULTS (%) WITH SAR AND TERRAIN INPUT AT INFERENCE. KD METHODS SHARE THE STUDENT, INITIALIZATION, AND FROZEN OPTICAL–SAR TEACHER.

Method	AP	IoU	F1	IoU Ind.	IoU Dep.
U-Net [6]	72.91	52.20	68.59	63.24	46.03
Vanilla KD [8]	70.89	53.45	69.66	68.52	44.97
AT [12]	76.63	53.12	69.39	61.22	48.63
Multi-to-Uni [10]	74.92	53.28	69.52	63.14	47.88
ACAM-KD [11]	74.54	54.59	70.63	66.09	48.45
PKDOT [15]	74.75	55.74	71.58	**70.52**	47.80
DIS2 [9]	75.85	55.97	71.77	69.71	48.06
Ours	**77.59**	**56.45**	**72.17**	69.46	**49.88**
Optical–SAR teacher	85.09	70.38	82.62	81.90	64.88

and the same frozen teacher (70.38% test IoU). We report AP and pixel-pooled IoU and F1 over the test set, and IoU on Ind. and Dep.

B. Main Comparison

Table I compares distillation methods that transfer outputs (Vanilla KD), attention or features (AT, Multi-to-Uni, ACAM-KD, DIS2), and pairwise relations (PKDOT). With identical deployment input, Ours improves IoU by 4.25 points and AP by 4.68 points over the U-Net, and gives the highest pooled AP, IoU, and F1 among the distillation methods, 0.48 IoU points above DIS2 and 0.96 AP points above the best competing AP (AT). The gain comes mainly from recall (70.3% versus 58.4%), at lower precision (74.2% versus 83.1%). On the independent events, PKDOT and DIS2 remain 1.06 and 0.25 points higher; Ours is highest on the dependent events. Output distillation gives the lowest AP, consistent with a teacher that reproduces the training labels.

C. Why Pairwise Relations Transfer Poorly

For every training and validation region we take the 30 regions of other events that the teacher finds least similar and measure the share that a frozen student also ranks among its own 30 least similar. Across 13 students trained with the same recipe, this agreement correlates negatively with test IoU ($r = -0.70$; Fig. 2). The best students agree with the teacher less than the U-Net does, whereas losses that enforce pair correspondence raise the agreement to 0.68–0.82 and test IoU falls to 52–53%. Pairwise teacher relations therefore carry structure tied to the training events; the distribution of cross-event similarities is what transfers.

D. Ablation Study

Table II isolates each component. L_{CEW} alone raises IoU from 52.20% to 55.57%. The shared prototypes alone leave IoU unchanged (52.07%) but raise AP to 77.99%: they improve the ranking of landslide pixels without changing the decisions at threshold 0.5. Combined with L_{CEW}, this ranking turns into detections and IoU reaches 56.45%, mainly through recall. Applying the relation loss also at the 64×64 stage lowers IoU to 55.22% and independent-event IoU from 70.84% to 64.85%, whereas sharing one prototype bank across

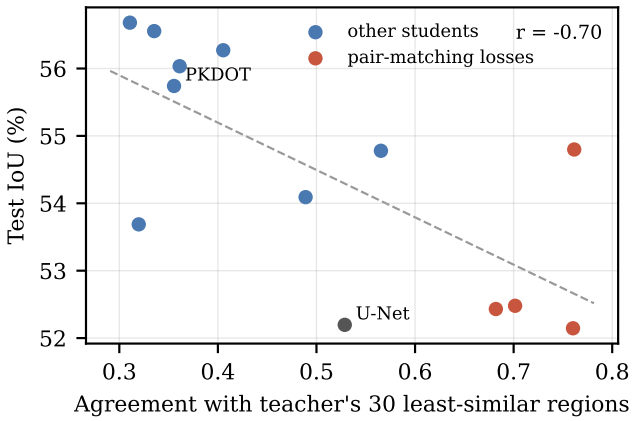


Fig. 2. Teacher–student agreement on the teacher's 30 least similar cross-event regions (validation patches) versus test IoU for students trained with the same recipe.

TABLE II
ABLATION (%). ALL ROWS USE THE SAME U-NET, INITIALIZATION, TEACHER, AND RECIPE.

Configuration	AP	IoU
U-Net	72.91	52.20
+ shared prototypes, three stages (no optical)	77.99	52.07
+ L_{CEW} (no prototypes)	76.41	55.57
+ prototypes at stage 2 + L_{CEW}	77.58	56.10
+ stage-wise prototypes, L_{CEW} at stages 2–3	75.51	55.22
+ shared prototypes, three stages + L_{CEW} (Ours)	77.59	56.45
Relation target, prototypes at stage 2		
Pairwise, rank-stratified partners	72.55	52.43
Pairwise, soft correspondence (KL)	72.43	52.48
Distribution, Wasserstein	77.58	56.10
Relation source, PKDOT loss		
Label-defined relations	74.96	54.09
Optical-teacher relations	74.75	55.74
Optical, cross-event pairs only	77.24	56.27

three stages while constraining only stage 2 gives the best result. With the same prototypes, losses that impose pair correspondence fall to 52.4–52.5% IoU, and label-defined relations trail optical-teacher relations by 1.65 points.

E. Deployment Cost

The three prototype modules add 0.14 M parameters and 0.36 G multiply–accumulate operations (MACs) to the 31.05 M-parameter, 12.07 GMAC U-Net per 128×128 patch (about 3%). The teacher and L_{CEW} are used only in training, so inference uses SAR and terrain alone.

IV. CONCLUSION

For event-grouped landslide inventories, an optical teacher transfers best to a SAR student through the distribution of its cross-event relations rather than through outputs, features, or pairwise relations. Matching this distribution with a closed-form Wasserstein loss and storing it in prototypes shared across events and scales improves SAR-only landslide mapping at about 3% additional deployment cost. All scores are single training runs; repeating one configuration with

identical settings changed test IoU by 0.65 points because of nondeterministic GPU kernels, so differences below this level should be read with caution. The method gains recall at some cost in precision and does not lead on the independent events.

REFERENCES

[1] A. C. Mondini, F. Guzzetti, K.-T. Chang, O. Monserrat, T. R. Martha, and A. Manconi, "Landslide failures detection and mapping using synthetic aperture radar: Past, present and future," *Earth-Science Reviews*, vol. 216, p. 103574, 2021.

[2] A. L. Handwerger, M.-H. Huang, S. Y. Jones, P. Amatya, H. R. Kerner, and D. B. Kirschbaum, "Generating landslide density heatmaps for rapid detection using open-access satellite radar data in Google Earth Engine," *Natural Hazards and Earth System Sciences*, vol. 22, no. 3, pp. 753–773, 2022.

[3] P. Amatya, C. Scheip, A. Déprez, J.-P. Malet, S. L. Slaughter, A. L. Handwerger, R. Emberson, D. Kirschbaum, J. Jean-Baptiste, M.-H. Huang, M. K. Clark, D. Zekkos, J.-R. Huang, F. Pacini, and E. Boissier, "Learnings from rapid response efforts to remotely detect landslides triggered by the August 2021 Nippes earthquake and Tropical Storm Grace in Haiti," *Natural Hazards*, vol. 118, no. 3, pp. 2337–2375, 2023.

[4] L. Nava, O. Monserrat, and F. Catani, "Improving landslide detection on SAR data through deep learning," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.

[5] L. Nava, K. Bhuyan, S. R. Meena, O. Monserrat, and F. Catani, "Rapid mapping of landslides on SAR data by attention U-Net," *Remote Sensing*, vol. 14, no. 6, p. 1449, 2022.

[6] A. Orynbaikyzy, F. Albrecht, W. Yao, M. Motagh, W. Wang, S. Martinis, and S. Plank, "Landslide mapping with deep learning: the role of pre-/post-event SAR features and multi-sensor data fusion," *GIScience & Remote Sensing*, vol. 62, no. 1, 2025.

[7] D. Lopez-Paz, L. Bottou, B. Schölkopf, and V. Vapnik, "Unifying distillation and privileged information," in *International Conference on Learning Representations*, 2016.

[8] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, 2015.

[9] N. Kieu, K. Nguyen, A. Wiliem, C. Fookes, and S. Sridharan, "DIS2: Disentanglement meets distillation with classwise attention for robust remote sensing segmentation under missing modalities," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision Workshops*, 2026, pp. 1373–1381.

[10] B. Sun, T. Hu, F. Zhang, L. Fang, and S. Li, "Multi-to-uni modality knowledge distillation for remote sensing image semantic segmentation," *Information Fusion*, vol. 138, p. 104696, Feb. 2027.

[11] Q. Lan and Q. Tian, "ACAM-KD: Adaptive and cooperative attention masking for knowledge distillation," in *Proc. IEEE/CVF Int. Conf. Computer Vision*, 2025, pp. 3957–3966.

[12] S. Zagoruyko and N. Komodakis, "Paying more attention to attention: Improving the performance of convolutional neural networks via attention transfer," in *International Conference on Learning Representations*, 2017.

[13] W. Park, D. Kim, Y. Lu, and M. Cho, "Relational knowledge distillation," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition*, 2019, pp. 3967–3976.

[14] F. Tung and G. Mori, "Similarity-preserving knowledge distillation," in *Proc. IEEE/CVF Int. Conf. Computer Vision*, 2019, pp. 1365–1374.

[15] M. H. Aslam, M. O. Zeeshan, S. Belharbi, M. Pedersoli, A. L. Koerich, S. Bacon, and E. Granger, "Distilling privileged multimodal information for expression recognition using optimal transport," in *IEEE International Conference on Automatic Face and Gesture Recognition (FG)*, 2024.

[16] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2015, pp. 234–241.